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Lab 11 – Data Structures with AI: Implementing Fundamental Structures

Lab Objectives

- Use AI to assist in designing and implementing fundamental data structures in Python.
- Learn how to prompt AI for structure creation, optimization, and documentation.
- Improve understanding of Lists, Stacks, Queues, Linked Lists, Trees, Graphs, and Hash Tables.
- Enhance code quality with AI-generated comments and performance suggestions.

Task Description #1 – Stack Implementation

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample Input Code:

```
class Stack:
```

```
    pass
```

Expected Output:

- A functional stack implementation with all required methods and docstrings.

PROMPT:

#Generate a simple and clean Python Stack class using a list with methods push, pop, peek, and is_empty, handle empty stack safely, include short student-style docstrings, keep code readable for a college lab record, and show a small example usage.

CODE:

```
class Stack:
```

```
    """A simple Stack class using a list to store elements."""
```

```
    def __init__(self):
```

```

        """Initialize an empty stack."""
        self.stack = []

def push(self, item):
    """Add an item to the top of the stack."""
    self.stack.append(item)

def pop(self):
    """Remove and return the item at the top of the stack.
    Returns None if the stack is empty."""
    if not self.is_empty():
        return self.stack.pop()
    else:
        print("Stack is empty. Cannot pop.")
        return None

def peek(self):
    """Return the item at the top of the stack without removing it.
    Returns None if the stack is empty."""
    if not self.is_empty():
        return self.stack[-1]
    else:
        print("Stack is empty. Cannot peek.")
        return None

def is_empty(self):
    """Check if the stack is empty."""
    return len(self.stack) == 0
# Example usage
if __name__ == "__main__":
    my_stack = Stack()
    my_stack.push(1)
    my_stack.push(2)
    my_stack.push(3)

    print("Top item:", my_stack.peek()) # Output: Top item: 3
    print("Popped item:", my_stack.pop()) # Output: Popped item: 3
    print("Is stack empty?", my_stack.is_empty()) # Output: Is stack empty? False
    print("Popped item:", my_stack.pop()) # Output: Popped item: 2
    print("Popped item:", my_stack.pop()) # Output: Popped item: 1
    print("Is stack empty?", my_stack.is_empty()) # Output: Is stack empty? True
    print("Popped item from empty stack:", my_stack.pop()) # Output: Stack is empty.
    Cannot pop. Popped item from empty stack: None
    print("Peek item from empty stack:", my_stack.peek()) # Output: Stack is empty. Cannot
    peek. Peek item from empty stack: None

```

OUTPUT:

Top item: 3
 Popped item: 3
 Is stack empty? False

Popped item: 2
Popped item: 1
Is stack empty? True
Stack is empty. Cannot pop.
Popped item from empty stack: None
Stack is empty. Cannot peek.
Peek item from empty stack: None

```
A11.1.py > ...
2 class Stack:
22     def peek(self):
25         if not self.is_empty():
26             return self.stack[-1]
27         else:
28             print("Stack is empty. Cannot peek.")
29             return None
30
31     def is_empty(self):
32         """Check if the stack is empty."""
33         return len(self.stack) == 0
34
35 # Example usage
36 if __name__ == "__main__":
37     my_stack = Stack()
38     my_stack.push(1)
39     my_stack.push(2)
40     my_stack.push(3)
41
42     print("Top item:", my_stack.peek()) # Output: Top item: 3
43     print("Popped item:", my_stack.pop()) # Output: Popped item: 3
44     print("Is stack empty?", my_stack.is_empty()) # Output: Is stack empty? False
45     print("Popped item:", my_stack.pop()) # Output: Popped item: 2
46     print("Popped item:", my_stack.pop()) # Output: Popped item: 1
47     print("Is stack empty?", my_stack.is_empty()) # Output: Is stack empty? True
48     print("Popped item from empty stack:", my_stack.pop()) # Output: Stack is empty. Cannot pop. Popped item from empty stack: None
49     print("Peek item from empty stack:", my_stack.peek()) # Output: Stack is empty. Cannot peek. Peek item from empty stack: None
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
Top item: 3
Popped item: 3
Is stack empty? False
Popped item: 2
Popped item: 1
Is stack empty? True
Stack is empty. Cannot pop.
Popped item from empty stack: None
Stack is empty. Cannot peek.
Peek item from empty stack: None
saivenkatesh@Sais-MacBook-Air AI Coding %
```

Task Description #2 – Queue Implementation

Task: Use AI to implement a Queue using Python lists.

Sample Input Code:

class Queue:

pass

Expected Output:

- FIFO-based queue class with enqueue, dequeue, peek, and size methods.

PROMPT:

#Generate a simple and clean Python Queue class using a list with FIFO behavior and methods enqueue, dequeue, peek, and size, handle empty queue safely, include short natural student-style docstrings, keep the code clear and

suitable for a college lab record, and provide a small example usage.

CODE:

```
class Queue:
    """A simple Queue class using a list to store elements with
    FIFO behavior."""

    def __init__(self):
        """Initialize an empty queue."""
        self.queue = []

    def enqueue(self, item):
        """Add an item to the end of the queue."""
        self.queue.append(item)

    def dequeue(self):
        """Remove and return the item at the front of the queue.
        Returns None if the queue is empty."""
        if not self.is_empty():
            return self.queue.pop(0)
        else:
            print("Queue is empty. Cannot dequeue.")
            return None

    def peek(self):
        """Return the item at the front of the queue without
        removing it.
        Returns None if the queue is empty."""
        if not self.is_empty():
            return self.queue[0]
        else:
            print("Queue is empty. Cannot peek.")
            return None

    def size(self):
        """Return the number of items in the queue."""
        return len(self.queue)

    def is_empty(self):
        """Check if the queue is empty."""
        return len(self.queue) == 0

# Example usage
if __name__ == "__main__":
    my_queue = Queue()
    my_queue.enqueue(1)
    my_queue.enqueue(2)
    my_queue.enqueue(3)
```

```

    print("Front item:", my_queue.peek()) # Output: Front item: 1
    print("Dequeued item:", my_queue.dequeue()) # Output:
Dequeued item: 1
    print("Queue size:", my_queue.size()) # Output: Queue size: 2
    print("Dequeued item:", my_queue.dequeue()) # Output:
Dequeued item: 2
    print("Dequeued item:", my_queue.dequeue()) # Output:
Dequeued item: 3
    print("Is queue empty?", my_queue.is_empty()) # Output: Is
queue empty? True
    print("Dequeued item from empty queue:", my_queue.dequeue())
# Output: Queue is empty. Cannot dequeue. Dequeued item from empty
queue: None
    print("Peek item from empty queue:", my_queue.peek()) #
Output: Queue is empty. Cannot peek. Peek item from empty queue:
None

```

OUTPUT:

```

Front item: 1
Dequeued item: 1
Queue size: 2
Dequeued item: 2
Dequeued item: 3
Is queue empty? True
Queue is empty. Cannot dequeue.
Dequeued item from empty queue: None
Queue is empty. Cannot peek.
Peek item from empty queue: None

```

Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and display methods.

Sample Input Code:

```
class Node:
```

```
    pass
```

```
class LinkedList:
```

```
    pass
```

Expected Output:

- A working linked list implementation with clear method documentation.

A11.1.py > ...

```
55 class Queue:
86     return len(self.queue)
87
88     def is_empty(self):
89         """Check if the queue is empty."""
90         return len(self.queue) == 0
91 # Example usage
92 if __name__ == "__main__":
93     my_queue = Queue()
94     my_queue.enqueue(1)
95     my_queue.enqueue(2)
96     my_queue.enqueue(3)
97
98     print("Front item:", my_queue.peek()) # Output: Front item: 1
99     print("Dequeued item:", my_queue.dequeue()) # Output: Dequeued item: 1
100    print("Queue size:", my_queue.size()) # Output: Queue size: 2
101    print("Dequeued item:", my_queue.dequeue()) # Output: Dequeued item: 2
102    print("Dequeued item:", my_queue.dequeue()) # Output: Dequeued item: 3
103    print("Is queue empty?", my_queue.is_empty()) # Output: Is queue empty? True
104    print("Dequeued item from empty queue:", my_queue.dequeue()) # Output: Queue is empty. Cannot dequeue. Dequeued item from empty queue: None
105    print("Peek item from empty queue:", my_queue.peek()) # Output: Queue is empty. Cannot peek. Peek item from empty queue: None
106
```

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```
/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
Front item: 1
Dequeued item: 1
Queue size: 2
Dequeued item: 2
Dequeued item: 3
Is queue empty? True
Queue is empty. Cannot dequeue.
Dequeued item from empty queue: None
Queue is empty. Cannot peek.
Peek item from empty queue: None
saivenkatesh@Sais-MacBook-Air AI Coding %
```

PROMPT:

#Generate a simple Python implementation of a Singly Linked List with a Node class and LinkedList class, include insert at end and display methods, use clear beginner-friendly docstrings and comments, avoid complex optimizations, keep the code clean for a college lab record, and provide a small example usage.

CODE:

```
class Node:
```

```
    """A Node class to represent each element in the linked list."""
```

```
    def __init__(self, data):
        """Initialize a node with data and a pointer to the next node."""
        self.data = data # Store the data
        self.next = None # Initialize the next pointer to None
```

```
class LinkedList:
```

```
    """A LinkedList class to manage the linked list operations."""
```

```
    def __init__(self):
        """Initialize an empty linked list."""
```

```

        self.head = None # Start with an empty list

    def insert_at_end(self, data):
        """Insert a new node with the given data at the end of the
list."""
        new_node = Node(data) # Create a new node
        if self.head is None: # If the list is empty, set the new
node as head
            self.head = new_node
            return
        last_node = self.head # Start from the head
        while last_node.next: # Traverse to the end of the list
            last_node = last_node.next
        last_node.next = new_node # Link the last node to the new
node

    def display(self):
        """Display the contents of the linked list."""
        current_node = self.head # Start from the head
        while current_node: # Traverse through the list
            print(current_node.data, end=' -> ') # Print current
node's data
            current_node = current_node.next # Move to the next
node
        print("None") # Indicate the end of the list
# Example usage
if __name__ == "__main__":
    my_list = LinkedList()
    my_list.insert_at_end(1)
    my_list.insert_at_end(2)
    my_list.insert_at_end(3)

    print("Linked List contents:")
    my_list.display() # Output: Linked List contents: 1 -> 2 -> 3
-> None

```

OUTPUT:

Linked List contents:
1 -> 2 -> 3 -> None

Task Description #4 – Binary Search Tree (BST)

Task: Use AI to create a BST with insert and in-order traversal methods.

Sample Input Code:

```
class BST:
```

```
    pass
```

Expected Output:

- BST implementation with recursive insert and traversal methods.

PROMPT:

#Generate a simple Python implementation of a Binary Search Tree with recursive insert and in-order traversal methods, use clear beginner-friendly docstrings, keep the code clean and easy for a college lab record, and include an example inserting values and printing them in sorted order.

CODE:

```
class TreeNode:
    """A TreeNode class to represent each node in the binary
    search tree."""

    def __init__(self, data):
        """Initialize a tree node with data and pointers to left
        and right children."""
        self.data = data # Store the data
        self.left = None # Initialize left child pointer
        self.right = None # Initialize right child pointer

class BinarySearchTree:
    """A BinarySearchTree class to manage the binary search tree
    operations."""

    def __init__(self):
        """Initialize an empty binary search tree."""
        self.root = None # Start with an empty tree

    def insert(self, data):
        """Insert a new node with the given data into the binary
        search tree."""
        if self.root is None: # If the tree is empty, set the new
            node as root
            self.root = TreeNode(data)
        else:
            self._insert_recursive(self.root, data) # Insert
            recursively

    def _insert_recursive(self, node, data):
        """Helper method to insert a new node recursively."""
        if data < node.data: # If data is less than current
            node's data
            if node.left is None: # If left child is empty,
                insert here
                node.left = TreeNode(data)
            else:
                self._insert_recursive(node.left, data) # Recur
            left
```



```

        else: # If data is greater than or equal to current
node's data
            if node.right is None: # If right child is empty,
insert here
                node.right = TreeNode(data)
            else:
                self._insert_recursive(node.right, data) # Recur
right

    def in_order_traversal(self):
        """Perform in-order traversal and return a list of values
in sorted order."""
        return self._in_order_recursive(self.root)

    def _in_order_recursive(self, node):
        """Helper method to perform in-order traversal
recursively."""
        result = []
        if node:
            result.extend(self._in_order_recursive(node.left)) #
Traverse left subtree
            result.append(node.data) # Visit current node
            result.extend(self._in_order_recursive(node.right)) #
Traverse right subtree
        return result

# Example usage
if __name__ == "__main__":
    bst = BinarySearchTree()
    bst.insert(5)
    bst.insert(3)
    bst.insert(7)
    bst.insert(2)
    bst.insert(4)
    bst.insert(6)
    bst.insert(8)

    print("In-order traversal (sorted order):",
bst.in_order_traversal()) # Output: In-order traversal (sorted
order): [2, 3, 4, 5, 6, 7, 8]

```

OUTPUT:

In-order traversal (sorted order): [2, 3, 4, 5, 6, 7, 8]

Task Description #5 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Sample Input Code:

```
class HashTable:
```

```
    pass
```

```
A11.1.py > ...
164 class BinarySearchTree:
195     def _in_order_recursive(self, node):
200         result.append(node.data) # Visit current node
201         result.extend(self._in_order_recursive(node.right)) # Traverse right subtree
202         return result
203 # Example usage
204 if __name__ == "__main__":
205     bst = BinarySearchTree()
206     bst.insert(5)
207     bst.insert(3)
208     bst.insert(7)
209     bst.insert(2)
210     bst.insert(4)
211     bst.insert(6)
212     bst.insert(8)
213
214     print("In-order traversal (sorted order):", bst.in_order_traversal()) # Output: In-order traversal (sorted order): [2, 3, 4, 5, 6, 7, 8]
215
```

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```
/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
In-order traversal (sorted order): [2, 3, 4, 5, 6, 7, 8]
saivenkatesh@Sais-MacBook-Air AI Coding %
```

Expected Output:

- Collision handling using chaining, with well-commented methods.

PROMPT:

#Generate a simple Python implementation of a Hash Table using chaining for collision handling with a list of buckets, include insert, search, and delete methods, use a basic hash function, write clear beginner-friendly comments and docstrings, keep the code clean for a college lab record, and provide a small demo example.

CODE:

```
class HashTable:
    """A simple Hash Table implementation using chaining for
    collision handling."""

    def __init__(self, size=10):
        """Initialize the hash table with a specified number of
        buckets."""
        self.size = size # Set the size of the hash table
        self.table = [[] for _ in range(size)] # Create a list of
        empty buckets

    def _hash(self, key):
```

```

        """Basic hash function to compute the index for a given
key."""
        return hash(key) % self.size # Compute hash and map it to
the table size

    def insert(self, key, value):
        """Insert a key-value pair into the hash table."""
        index = self._hash(key) # Get the index for the key
        bucket = self.table[index] # Get the bucket at that index
        for i, (k, v) in enumerate(bucket): # Check if the key
already exists
            if k == key:
                bucket[i] = (key, value) # Update value if key
exists
            return
        bucket.append((key, value)) # Add new key-value pair if
key does not exist

    def search(self, key):
        """Search for a value by its key. Returns None if not
found."""
        index = self._hash(key) # Get the index for the key
        bucket = self.table[index] # Get the bucket at that index
        for k, v in bucket: # Search through the bucket
            if k == key:
                return v # Return value if key is found
        return None # Return None if key is not found

    def delete(self, key):
        """Delete a key-value pair from the hash table. Returns
True if deleted, False if not found."""
        index = self._hash(key) # Get the index for the key
        bucket = self.table[index] # Get the bucket at that index
        for i, (k, v) in enumerate(bucket): # Search through the
bucket
            if k == key:
                del bucket[i] # Delete the key-value pair if
found
            return True
        return False # Return False if key is not found

# Example usage
if __name__ == "__main__":
    my_hash_table = HashTable()
    my_hash_table.insert("name", "Alice")
    my_hash_table.insert("age", 30)
    my_hash_table.insert("city", "New York")

    print("Search for 'name':", my_hash_table.search("name")) #
Output: Search for 'name': Alice
    print("Search for 'age':", my_hash_table.search("age")) #
Output: Search for 'age': 30

```

```

        print("Search for 'country':",
my_hash_table.search("country")) # Output: Search for 'country':
None

        print("Delete 'age':", my_hash_table.delete("age")) # Output:
Delete 'age': True
        print("Search for 'age' after deletion:",
my_hash_table.search("age")) # Output: Search for 'age' after
deletion: None

```

OUTPUT:

Search for 'name': Alice
 Search for 'age': 30
 Search for 'country': None
 Delete 'age': True
 Search for 'age' after deletion: None

```

A11.1.py > ...
220 class HashTable:
251     def delete(self, key):
252         for i, (k, v) in enumerate(bucket): # Search through the bucket
253             if k == key:
254                 del bucket[i] # Delete the key-value pair if found
255                 return True
256             return False # Return False if key is not found
257
258 # Example usage
259
260 if __name__ == "__main__":
261     my_hash_table = HashTable()
262     my_hash_table.insert("name", "Alice")
263     my_hash_table.insert("age", 30)
264     my_hash_table.insert("city", "New York")
265
266     print("Search for 'name':", my_hash_table.search("name")) # Output: Search for 'name': Alice
267     print("Search for 'age':", my_hash_table.search("age")) # Output: Search for 'age': 30
268     print("Search for 'country':", my_hash_table.search("country")) # Output: Search for 'country': None
269
270     print("Delete 'age':", my_hash_table.delete("age")) # Output: Delete 'age': True
271     print("Search for 'age' after deletion:", my_hash_table.search("age")) # Output: Search for 'age' after deletion: None
272
273

```

PROBLEMS

OUTPUT

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```

/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
In-order traversal (sorted order): [2, 3, 4, 5, 6, 7, 8]
saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
Search for 'name': Alice
Search for 'age': 30
Search for 'country': None
Delete 'age': True
Search for 'age' after deletion: None
saivenkatesh@Sais-MacBook-Air AI Coding %

```

Task Description #6 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

```
class Graph:
```

pass

Expected Output:

- Graph with methods to add vertices, add edges, and display connections.

PROMPT:

#Generate a simple and clean Python Graph class using an adjacency list for an undirected graph, include add_vertex, add_edge, and display methods, write clear beginner-friendly docstrings, keep the implementation readable for a college lab record, and include a small example showing vertex connections.

CODE:

```
class Graph:
    """A simple Graph class using an adjacency list for an
    undirected graph."""

    def __init__(self):
        """Initialize an empty graph."""
        self.graph = {} # Use a dictionary to store the adjacency
list

    def add_vertex(self, vertex):
        """Add a vertex to the graph."""
        if vertex not in self.graph:
            self.graph[vertex] = [] # Initialize the vertex with
an empty list of neighbors

    def add_edge(self, vertex1, vertex2):
        """Add an undirected edge between two vertices in the
graph."""
        if vertex1 not in self.graph:
            self.add_vertex(vertex1) # Add vertex1 if it doesn't
exist
        if vertex2 not in self.graph:
            self.add_vertex(vertex2) # Add vertex2 if it doesn't
exist
        self.graph[vertex1].append(vertex2) # Add vertex2 to
vertex1's neighbors
        self.graph[vertex2].append(vertex1) # Add vertex1 to
vertex2's neighbors

    def display(self):
        """Display the adjacency list of the graph."""
        for vertex, neighbors in self.graph.items():
            print(f"{vertex}: {' '.join(neighbors)}") # Print
each vertex and its neighbors
# Example usage
if __name__ == "__main__":
    my_graph = Graph()
```

```

my_graph.add_edge("A", "B")
my_graph.add_edge("A", "C")
my_graph.add_edge("B", "D")
my_graph.add_edge("C", "D")
my_graph.add_edge("D", "E")

print("Graph adjacency list:")
my_graph.display() # Output: Graph adjacency list: A: B, C B:
A, D C: A, D D: B, C, E E: D

```

OUTPUT:

Graph adjacency list:

A: B, C

B: A, D

C: A, D

D: B, C, E

E: D

```

A11.1.py > ...
275 class Graph:
295
296     def display(self):
297         """Display the adjacency list of the graph."""
298         for vertex, neighbors in self.graph.items():
299             print(f"{vertex}: {' '.join(neighbors)}") # Print each vertex and its neighbors
300 # Example usage
301 if __name__ == "__main__":
302     my_graph = Graph()
303     my_graph.add_edge("A", "B")
304     my_graph.add_edge("A", "C")
305     my_graph.add_edge("B", "D")
306     my_graph.add_edge("C", "D")
307     my_graph.add_edge("D", "E")
308
309     print("Graph adjacency list:")
310     my_graph.display() # Output: Graph adjacency list: A: B, C B: A, D C: A, D D: B, C, E E: D
311

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE Python + v

```

/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"
Graph adjacency list:
A: B, C
B: A, D
C: A, D
D: B, C, E
E: D
saivenkatesh@Sais-MacBook-Air AI Coding %

```

Task Description #7 – Priority Queue

Task: Use AI to implement a priority queue using Python's heapq module.

Sample Input Code:

```
class PriorityQueue:
```

```
    pass
```

Expected Output:

- Implementation with enqueue (priority), dequeue (highest priority), and display methods.

PROMPT:

#Generate a simple Python Priority Queue using the heapq module with enqueue(value, priority), dequeue returning the highest priority item, and display methods, include short student-style docstrings, keep the code clean and readable for a college lab record, and provide a small example usage.

CODE:

```
import heapq
class PriorityQueue:
    """A simple Priority Queue implementation using the heapq
    module."""

    def __init__(self):
        """Initialize an empty priority queue."""
        self.queue = [] # Use a list to store the priority queue

    def enqueue(self, value, priority):
        """Add an item to the priority queue with a given
        priority."""
        heapq.heappush(self.queue, (priority, value)) # Push a
        tuple of (priority, value)

    def dequeue(self):
        """Remove and return the item with the highest priority.
        Returns None if the queue is empty."""
        if not self.is_empty():
            return heapq.heappop(self.queue)[1] # Pop the item
            with the highest priority
        else:
            print("Priority Queue is empty. Cannot dequeue.")
            return None

    def display(self):
        """Display the contents of the priority queue."""
        print("Priority Queue contents:")
        for priority, value in sorted(self.queue): # Sort by
            priority for display
            print(f"Value: {value}, Priority: {priority}")

    def is_empty(self):
```

```

        """Check if the priority queue is empty."""
        return len(self.queue) == 0
# Example usage
if __name__ == "__main__":
    pq = PriorityQueue()
    pq.enqueue("Task 1", priority=2)
    pq.enqueue("Task 2", priority=1)
    pq.enqueue("Task 3", priority=3)

    pq.display() # Output: Priority Queue contents: Value: Task
2, Priority: 1 Value: Task 1, Priority: 2 Value: Task 3, Priority:
3

    print("Dequeued item:", pq.dequeue()) # Output: Dequeued
item: Task 2
    print("Dequeued item:", pq.dequeue()) # Output: Dequeued
item: Task 1
    print("Dequeued item:", pq.dequeue()) # Output: Dequeued
item: Task 3
    print("Dequeued item from empty queue:", pq.dequeue()) #
Output: Priority Queue is empty. Cannot dequeue. Dequeued item
from empty queue: None

```

OUTPUT:

Priority Queue contents:
 Value: Task 2, Priority: 1
 Value: Task 1, Priority: 2
 Value: Task 3, Priority: 3
 Dequeued item: Task 2
 Dequeued item: Task 1
 Dequeued item: Task 3
 Priority Queue is empty. Cannot dequeue.
 Dequeued item from empty queue: None

Task Description #8 – Deque

Task: Use AI to implement a double-ended queue using collections.deque.

Sample Input Code:

```
class DequeDS:
```

```
    pass
```

Expected Output:

- Insert and remove from both ends with docstrings.

PROMPT:

```
#Generate a simple Python Deque implementation using
collections.deque with insert_front, insert_rear, remove_front,
```



```

A11.1.py > ...
314 class PriorityQueue:
315     """Priority Queue implementation using heapq module"""
316     def __init__(self):
317         self.queue = []
318         return None
319
320 def display(self):
321     """Display the contents of the priority queue."""
322     print("Priority Queue contents:")
323     for priority, value in sorted(self.queue): # Sort by priority for display
324         print(f"Value: {value}, Priority: {priority}")
325
326 def is_empty(self):
327     """Check if the priority queue is empty."""
328     return len(self.queue) == 0
329
330 # Example usage
331 if __name__ == "__main__":
332     pq = PriorityQueue()
333     pq.enqueue("Task 1", priority=2)
334     pq.enqueue("Task 2", priority=1)
335     pq.enqueue("Task 3", priority=3)
336
337 pq.display() # Output: Priority Queue contents: Value: Task 2, Priority: 1 Value: Task 1, Priority: 2 Value: Task 3, Priority: 3
338
339 print("Dequeued item:", pq.dequeue()) # Output: Dequeued item: Task 2
340 print("Dequeued item:", pq.dequeue()) # Output: Dequeued item: Task 1
341 print("Dequeued item:", pq.dequeue()) # Output: Dequeued item: Task 3
342 print("Dequeued item from empty queue:", pq.dequeue()) # Output: Priority Queue is empty. Cannot dequeue. Dequeued item from empty queue: None

```

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS	POSTMAN CONSOLE
<pre> saivenkatesh@Saish-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py" saivenkatesh@Saish-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py" Priority Queue contents: Value: Task 2, Priority: 1 Value: Task 1, Priority: 2 Value: Task 3, Priority: 3 Dequeued item: Task 2 Dequeued item: Task 1 Dequeued item: Task 3 Priority Queue is empty. Cannot dequeue. Dequeued item from empty queue: None </pre>					

and remove_rear methods, include clear natural docstrings, keep the code clean and readable for a college lab record, and provide a small example usage.

CODE:

```

from collections import deque
class Deque:
    """A simple Deque (double-ended queue) implementation using collections.deque."""

    def __init__(self):
        """Initialize an empty deque."""
        self.deque = deque() # Use deque from the collections module

    def insert_front(self, item):
        """Insert an item at the front of the deque."""
        self.deque.appendleft(item) # Add item to the front

    def insert_rear(self, item):
        """Insert an item at the rear of the deque."""
        self.deque.append(item) # Add item to the rear

    def remove_front(self):
        """Remove and return the item at the front of the deque. Returns None if the deque is empty."""

```

```

        if not self.is_empty():
            return self.deque.popleft() # Remove and return item
from the front
        else:
            print("Deque is empty. Cannot remove from front.")
            return None

    def remove_rear(self):
        """Remove and return the item at the rear of the deque.
        Returns None if the deque is empty."""
        if not self.is_empty():
            return self.deque.pop() # Remove and return item from
the rear
        else:
            print("Deque is empty. Cannot remove from rear.")
            return None

    def is_empty(self):
        """Check if the deque is empty."""
        return len(self.deque) == 0
# Example usage
if __name__ == "__main__":
    my_deque = Deque()
    my_deque.insert_rear(1)
    my_deque.insert_rear(2)
    my_deque.insert_front(0)

    print("Removed from front:", my_deque.remove_front()) #
Output: Removed from front: 0
    print("Removed from rear:", my_deque.remove_rear()) # Output:
Removed from rear: 2
    print("Removed from rear:", my_deque.remove_rear()) # Output:
Removed from rear: 1
    print("Removed from empty deque:", my_deque.remove_front()) #
Output: Deque is empty. Cannot remove from front. Removed from
empty deque: None
    print("Removed from empty deque:", my_deque.remove_rear()) #
Output: Deque is empty. Cannot remove from rear. Removed from
empty deque: None

```

OUTPUT:

```

Removed from front: 0
Removed from rear: 2
Removed from rear: 1
Deque is empty. Cannot remove from front.
Removed from empty deque: None
Deque is empty. Cannot remove from rear.
Removed from empty deque: None

```

```
A11.1.py > ...
360 class Deque:
375     def remove_front(self):
382         return None
383
384     def remove_rear(self):
385         """Remove and return the item at the rear of the deque.
386         Returns None if the deque is empty."""
387         if not self.is_empty():
388             return self.deque.pop() # Remove and return item from the rear
389         else:
390             print("Deque is empty. Cannot remove from rear.")
391             return None
392
393     def is_empty(self):
394         """Check if the deque is empty."""
395         return len(self.deque) == 0
396
397 # Example usage
398 if __name__ == "__main__":
399     my_deque = Deque()
400     my_deque.insert_rear(1)
401     my_deque.insert_rear(2)
402     my_deque.insert_front(0)
403
404     print("Removed from front:", my_deque.remove_front()) # Output: Removed from front: 0
405     print("Removed from rear:", my_deque.remove_rear()) # Output: Removed from rear: 2
406     print("Removed from rear:", my_deque.remove_rear()) # Output: Removed from rear: 1
407     print("Removed from empty deque:", my_deque.remove_front()) # Output: Deque is empty. Cannot remove from front. Removed from empty deque: N
408     print("Removed from empty deque:", my_deque.remove_rear()) # Output: Deque is empty. Cannot remove from rear. Removed from empty deque: Non
```

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS	POSTMAN CONSOLE
	saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"				
	Priority Queue is empty. Cannot dequeue. Dequeued item from empty queue: None				
●	saivenkatesh@Sais-MacBook-Air AI Coding % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/A11.1.py"				
	Removed from front: 0 Removed from rear: 2 Removed from rear: 1 Deque is empty. Cannot remove from front. Removed from empty deque: None Deque is empty. Cannot remove from rear. Removed from empty deque: None				

Task Description #9 Real-Time Application Challenge – Choose the Right Data Structure

Scenario:

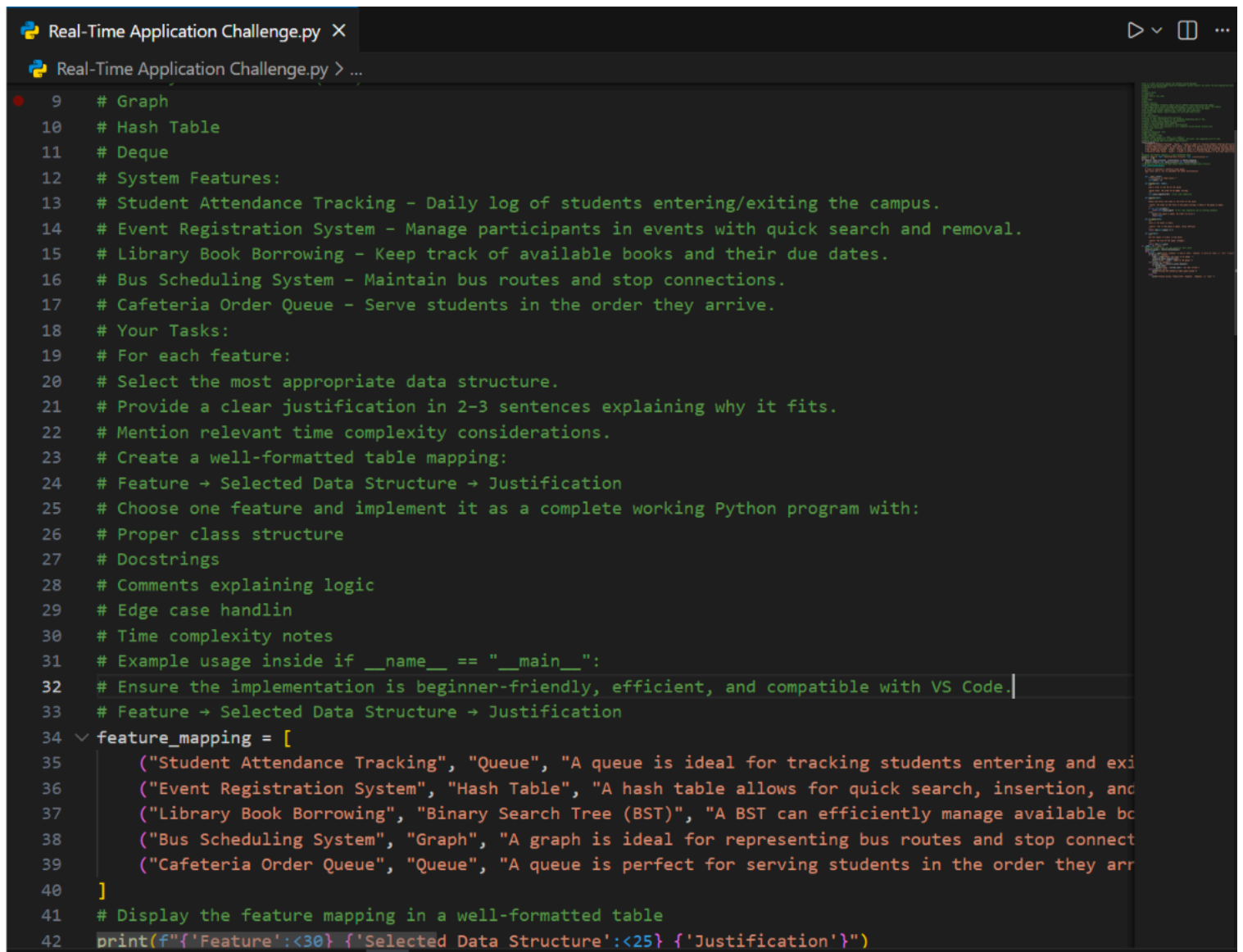
Your college wants to develop a Campus Resource Management System that handles:

1. Student Attendance Tracking – Daily log of students entering/exiting the campus.
2. Event Registration System – Manage participants in events with quick search and removal.
3. Library Book Borrowing – Keep track of available books and their due dates.
4. Bus Scheduling System – Maintain bus routes and stop connections.

5. Cafeteria Order Queue – Serve students in the order they arrive.

Student Task:

- For each feature, select the most appropriate data structure from the list below:
 - Stack
 - Queue
 - Priority Queue
 - Linked List



```
9 # Graph
10 # Hash Table
11 # Deque
12 # System Features:
13 # Student Attendance Tracking - Daily log of students entering/exiting the campus.
14 # Event Registration System - Manage participants in events with quick search and removal.
15 # Library Book Borrowing - Keep track of available books and their due dates.
16 # Bus Scheduling System - Maintain bus routes and stop connections.
17 # Cafeteria Order Queue - Serve students in the order they arrive.
18 # Your Tasks:
19 # For each feature:
20 # Select the most appropriate data structure.
21 # Provide a clear justification in 2-3 sentences explaining why it fits.
22 # Mention relevant time complexity considerations.
23 # Create a well-formatted table mapping:
24 # Feature → Selected Data Structure → Justification
25 # Choose one feature and implement it as a complete working Python program with:
26 # Proper class structure
27 # Docstrings
28 # Comments explaining logic
29 # Edge case handling
30 # Time complexity notes
31 # Example usage inside if __name__ == "__main__":
32 # Ensure the implementation is beginner-friendly, efficient, and compatible with VS Code.
33 # Feature → Selected Data Structure → Justification
34 feature_mapping = [
35     ("Student Attendance Tracking", "Queue", "A queue is ideal for tracking students entering and exiting the campus."),
36     ("Event Registration System", "Hash Table", "A hash table allows for quick search, insertion, and removal of participants."),
37     ("Library Book Borrowing", "Binary Search Tree (BST)", "A BST can efficiently manage available books and their due dates."),
38     ("Bus Scheduling System", "Graph", "A graph is ideal for representing bus routes and stop connections."),
39     ("Cafeteria Order Queue", "Queue", "A queue is perfect for serving students in the order they arrive.")
40 ]
41 # Display the feature mapping in a well-formatted table
42 print(f"{'Feature':<30} {'Selected Data Structure':<25} {'Justification':<40}")
```

- Binary Search Tree (BST)
- Graph
- Hash Table
- Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

```
Real-Time Application Challenge.py X
Real-Time Application Challenge.py > ...

40 ]
41 # Display the feature mapping in a well-formatted table
42 print(f"{'Feature':<30} {'Selected Data Structure':<25} {'Justification'}")
43 print("-" * 80)
44 for feature, data_structure, justification in feature_mapping:
45     print(f"{'feature':<30} {'data_structure':<25} {'justification'}")
46 # Implementing the Cafeteria Order Queue feature using a Queue data structure
47 class CafeteriaOrderQueue:
48     """
49     A class to represent a cafeteria order queue.
50     This class uses a list to implement the queue functionality.
51     """
52
53     def __init__(self):
54         """Initialize an empty queue."""
55         self.queue = []
56
57     def enqueue(self, order):
58         """
59         Add an order to the end of the queue.
60
61         :param order: The order to be added (string).
62         """
63         self.queue.append(order) # O(1) time complexity
64
65     def dequeue(self):
66         """
67         Remove and return the order at the front of the queue.
68
69         :return: The order at the front of the queue (string) or None if the queue is empty.
70         """
71         if not self.is_empty():
72             return self.queue.pop(0) # O(n) time complexity due to shifting elements
73         else:
```

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

```

47 class CafeteriaOrderQueue:
65     def dequeue(self):
69         :return: The order at the front of the queue (string) or None if the queue is empty.
70         """
71         if not self.is_empty():
72             return self.queue.pop(0) # O(n) time complexity due to shifting elements
73         else:
74             print("The queue is empty. No orders to serve.")
75             return None
76
77     def is_empty(self):
78         """
79         Check if the queue is empty.
80
81         :return: True if the queue is empty, False otherwise.
82         """
83         return len(self.queue) == 0
84
85     def size(self):
86         """
87         Get the number of orders in the queue.
88
89         :return: The size of the queue (integer).
90         """
91         return len(self.queue)
92 if __name__ == "__main__":
93     #give me dynamic input for the cafeteria order queue
94     cafeteria_queue = CafeteriaOrderQueue()
95     while True:
96         action = input("Enter 'enqueue' to add an order, 'dequeue' to serve an order, or 'exit' to qu
97         if action == 'enqueue':
98             order = input("Enter the order to be added: ")
99             cafeteria_queue.enqueue(order)
100             print(f"Order '{order}' added to the queue.")

```

```

47 class CafeteriaOrderQueue:
85     def size(self):
87         Get the number of orders in the queue.
88
89         :return: The size of the queue (integer).
90         """
91         return len(self.queue)
92 if __name__ == "__main__":
93     #give me dynamic input for the cafeteria order queue
94     cafeteria_queue = CafeteriaOrderQueue()
95     while True:
96         action = input("Enter 'enqueue' to add an order, 'dequeue' to serve an order, or 'exit' to qu
97         if action == 'enqueue':
98             order = input("Enter the order to be added: ")
99             cafeteria_queue.enqueue(order)
100             print(f"Order '{order}' added to the queue.")
101         elif action == 'dequeue':
102             served_order = cafeteria_queue.dequeue()
103             if served_order:
104                 print(f"Order '{served_order}' has been served.")
105         elif action == 'exit':
106             print("Exiting the cafeteria order queue system.")
107             break
108         else:
109             print("Invalid action. Please enter 'enqueue', 'dequeue', or 'exit'.")
110

```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + v [ ] [ ] ... | [ ] [ ] X

PS C:\Users\sonu\Desktop\AI_Coding> & C:/Python314/python.exe "c:/Users/sonu/Desktop/AI_Coding/Real-Time Application Challenge.py"
Feature                               Selected Data Structure   Justification
-----
Student Attendance Tracking           Queue                     A queue is ideal for tracking students entering and exiting the campus as it follows the First-In-First-Out (FIFO) principle, allowing for efficient logging of attendance. Time complexity for enqueue and dequeue operations is O(1).
Event Registration System              Hash Table                A hash table allows for quick search, insertion, and deletion of participants in events, making it suitable for managing registrations. Time complexity for search, insert, and delete operations is O(1) on average.
Library Book Borrowing                 Binary Search Tree (BST) A BST can efficiently manage available books and their due dates by allowing for sorted data storage and quick retrieval. Time complexity for search, insert, and delete operations is O(log n) on average.
Bus Scheduling System                  Graph                     A graph is ideal for representing bus routes and stop connections as it can model complex relationships between stops and routes. Time complexity for traversal operations is O(V + E), where V is the number of vertices (stops) and E is the number of edges (routes).
Cafeteria Order Queue                  Queue                     A queue is perfect for serving students in the order they arrive at the cafeteria, ensuring fairness. Time complexity for enqueue and dequeue operations is O(1).
Enter 'enqueue' to add an order, 'dequeue' to serve an order, or 'exit' to quit: enqueue
Enter the order to be added: 5 7 9 10
Order '5 7 9 10' added to the queue.
Enter 'enqueue' to add an order, 'dequeue' to serve an order, or 'exit' to quit: dequeue
Order '5 7 9 10' has been served.
Enter 'enqueue' to add an order, 'dequeue' to serve an order, or 'exit' to quit: exit
Exiting the cafeteria order queue system.
PS C:\Users\sonu\Desktop\AI_Coding>
```

Task Description #10: Smart E-Commerce Platform – Data Structure Challenge

An e-commerce company wants to build a Smart Online Shopping System with:

1. Shopping Cart Management – Add and remove products dynamically.
2. Order Processing System – Orders processed in the order they are placed.
3. Top-Selling Products Tracker – Products ranked by sales count.
4. Product Search Engine – Fast lookup of products using product ID.
5. Delivery Route Planning – Connect warehouses and delivery locations.

Student Task:

- For each feature, select the most appropriate data structure from the list below:
 - Stack
 - Queue
 - Priority Queue
 - Linked List

```

Smart E-Commerce Platform .py X
Smart E-Commerce Platform .py > ...
1  # Act as a data structures expert and system architect.
2  # Analyze the following Smart E-Commerce Platform requirements and select the most appropriate data s
3  # Available Data Structures:
4  # Stack
5  # Queue
6  # Priority Queue
7  # Linked List
8  # Binary Search Tree (BST)
9  # Graph
10 # Hash Table
11 # Deque
12 # System Features:
13 # Shopping Cart Management - Add and remove products dynamically.
14 # Order Processing System - Orders processed in the order they are placed.
15 # Top-Selling Products Tracker - Products ranked by sales count.
16 # Product Search Engine - Fast lookup of products using product ID.
17 # Delivery Route Planning - Connect warehouses and delivery locations.
18 # Required Tasks:
19 # For each feature:
20 # Select the most appropriate data structure.
21 # Provide a clear 2-3 sentence justification.
22 # Mention relevant time complexity considerations.
23 # Present the results in a well-formatted table:
24 # Feature → Selected Data Structure → Justification
25 # Choose one feature and implement it as a complete Python program with:
26 # Proper class-based structure
27 # Docstrings for class and methods
28 # Inline comments explaining logic
29 # Edge case handling
30 # Time complexity notes
31 # Example usage under if __name__ == "__main__":|
32 # Code compatible with VS Code
33 # Ensure the solution is technically accurate, beginner-friendly, and follows clean coding practices.
34 # Feature → Selected Data Structure → Justification

```

- Binary Search Tree (BST)
- Graph
- Hash Table
- Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.


```
Smart E-Commerce Platform .py X
Smart E-Commerce Platform .py > ...
33 # Ensure the solution is technically accurate, beginner-friendly, and follows clean coding practices.
34 # Feature → Selected Data Structure → Justification
35 features_data_structures = [
36     ("Shopping Cart Management", "Linked List", "A linked list allows for dynamic addition and removal of items in a shopping cart."),
37     ("Order Processing System", "Queue", "A queue is ideal for processing orders in the order they are received."),
38     ("Top-Selling Products Tracker", "Priority Queue", "A priority queue can efficiently keep track of top-selling products based on sales volume or profit margin."),
39     ("Product Search Engine", "Hash Table", "A hash table allows for fast lookup of products using product IDs or keywords as keys."),
40     ("Delivery Route Planning", "Graph", "A graph is suitable for modeling warehouses and delivery locations, enabling efficient route planning algorithms like Dijkstra's or A* search.")
41 ]
42 # Display the results in a well-formatted table
43 print(f"{'Feature':<30} {'Selected Data Structure':<25} {'Justification'}")
44 for feature, data_structure, justification in features_data_structures:
45     print(f"{'Feature':<30} {'Selected Data Structure':<25} {'Justification'}")
46 # Implementing the Shopping Cart Management feature using a Linked List
47 class Node:
48     """A node in the linked list representing a product in the shopping cart."""
49     def __init__(self, product_id, quantity):
50         self.product_id = product_id
51         self.quantity = quantity
52         self.next = None
53 class ShoppingCart:
54     """A shopping cart implemented as a linked list."""
55     def __init__(self):
56         self.head = None
57     def add_product(self, product_id, quantity):
58         """Add a product to the shopping cart."""
59         new_node = Node(product_id, quantity)
60         new_node.next = self.head
61         self.head = new_node
62     def remove_product(self, product_id):
63         """Remove a product from the shopping cart."""
64         current = self.head
65         previous = None
66         while current is not None:
```

- A functional Python program implementing the chosen feature with comments and docstrings.

```

Smart E-Commerce Platform.py X
Smart E-Commerce Platform.py > ...
53 class ShoppingCart:
54     self.head = new_node
62     def remove_product(self, product_id):
63         """Remove a product from the shopping cart."""
64         current = self.head
65         previous = None
66         while current is not None:
67             if current.product_id == product_id:
68                 if previous is None: # Removing the head node
69                     self.head = current.next
70                 else:
71                     previous.next = current.next
72                 return True # Product removed successfully
73                 previous = current
74                 current = current.next
75         return False # Product not found in the cart
76     def display_cart(self):
77         """Display the contents of the shopping cart."""
78         products = []
79         current = self.head
80         while current is not None:
81             products.append(f"Product ID: {current.product_id}, Quantity: {current.quantity}")
82             current = current.next
83         return products
84 if __name__ == "__main__":
85     #give me dynamic input for the shopping cart management
86     cart = ShoppingCart()
87     while True:
88         action = input("Enter 'add' to add a product, 'remove' to remove a product, 'display' to show
89         if action == 'add':
90             product_id = input("Enter product ID: ")
91             quantity = int(input("Enter quantity: "))
92             cart.add_product(product_id, quantity)
93             print(f"Added Product ID: {product_id} with Quantity: {quantity}")

```

```

Smart E-Commerce Platform.py X
Smart E-Commerce Platform.py > ...
53 class ShoppingCart:
76     def display_cart(self):
83         return products
84 if __name__ == "__main__":
85     #give me dynamic input for the shopping cart management
86     cart = ShoppingCart()
87     while True:
88         action = input("Enter 'add' to add a product, 'remove' to remove a product, 'display' to show
89         if action == 'add':
90             product_id = input("Enter product ID: ")
91             quantity = int(input("Enter quantity: "))
92             cart.add_product(product_id, quantity)
93             print(f"Added Product ID: {product_id} with Quantity: {quantity}")
94         elif action == 'remove':
95             product_id = input("Enter product ID to remove: ")
96             if cart.remove_product(product_id):
97                 print(f"Removed Product ID: {product_id}")
98             else:
99                 print(f"Product ID: {product_id} not found in the cart.")
100         elif action == 'display':
101             print("Shopping Cart Contents:")
102             for item in cart.display_cart():
103                 print(item)
104         elif action == 'exit':
105             print("Exiting the shopping cart management system.")
106             break
107         else:
108             print("Invalid action. Please try again.")
109

```

Feature	Selected Data Structure	Justification
Shopping Cart Management	Linked List	A linked list allows for dynamic addition and removal of products without needing to shift elements, making it efficient for managing a shopping cart. Time complexity for adding/removing products is $O(1)$.
Order Processing System	Queue	A queue is ideal for processing orders in the order they are received (FIFO). Time complexity for enqueue and dequeue operations is $O(1)$.
Top-Selling Products Tracker	Priority Queue	A priority queue can efficiently keep track of top-selling products by maintaining a sorted order based on sales count. Time complexity for insertion and retrieval is $O(\log n)$.
Product Search Engine	Hash Table	A hash table allows for fast lookup of products using product IDs with an average time complexity of $O(1)$ for search operations. It also handles collisions effectively.
Delivery Route Planning	Graph	A graph is suitable for modeling warehouses and delivery locations, allowing for efficient route planning using algorithms like Dijkstra's or A*. Time complexity depends on the algorithm used, typically $O(V + E)$ for Dijkstra's algorithm.
Enter 'add' to add a product, 'remove' to remove a product, 'display' to show cart contents, or 'exit' to quit: add		
Enter product ID: 567903		
Enter quantity: 300		
Added Product ID: 567903 with Quantity: 300		
Enter 'add' to add a product, 'remove' to remove a product, 'display' to show cart contents, or 'exit' to quit: display		